

Artificial intelligence in drug discovery – what it is, where we stand and the path forward

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Abstract

Artificial intelligence (AI) in drug discovery has attracted increasing interest over the past decade. It is now time for a critical review of progress in the field: where did we advance – and where are we yet to see impact – when it comes to what matters in drug discovery, which is to deliver safer and more efficacious medicines to patients faster? Although a wide variety of AI methods have been developed, applied and benchmarked, evidence of their clinically relevant impact is, so far, disappointingly limited. In this Perspective we discuss potential reasons, including an insufficient focus on clinical translation during model development, difficulties with applying AI algorithms on conditional life science data, and insufficient problem definitions and the resulting underspecification of computational models for real-world use cases. ‘Technology push’ compared with ‘science pull’ is also likely to be an underlying factor, as well as the substantial time required to operationalize technical capabilities into systems that are sufficiently scaled and accessible for users. We provide recommendations for the development of AI in drug discovery with the aim of increasing its translational relevance. For example, benchmarking studies of AI tools in drug discovery need to move on from model validation and instead focus on their ability to improve decision making.

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Introduction

The impact of artificial intelligence (AI) has already been profound in areas such as image^{1,2} and speech recognition³; for example, voice recognition is now built into many phones, offering remarkable accuracy. AI – by some definitions – has also been applied in drug research and development (R&D) over at least the past decade⁴, but thus far, its impact has been less pronounced. Two recent reviews of the field^{5,6} show that the large majority of projects in ‘AI-first’ drug discovery companies are in preclinical stages, with several dozen in phase I and phase II and only very few in phase III⁶. This as-yet modest achievement cannot be fully ascribed to the long duration of drug discovery projects, as many AI proponents have claimed these timelines will be meaningfully shortened. Although FDA approval rates showed an upward trend over the last several years⁷, there are multiple alternative explanations for this progress, including increasing investment in the development of biologic modalities that have higher average clinical success rates⁸ and superior intellectual property protection status⁹ than small-molecule drugs, and a growing contribution of rare disease approvals. In addition, although recent publications examining a small number of data points describe an apparent increase in success rates of projects in clinical phase I of up to 90% and attribute this to a potential impact of AI methods¹⁰, most of these projects are grounded in established disease biology and chemistry, reducing the likelihood of encountering safety issues the second (or a later time) around¹¹.

A very realistic, detailed viewpoint on the use of AI and machine learning (ML) in drug discovery has recently been published¹², and consortia reviews on model-informed drug discovery and development provide summaries of the current state of affairs¹³. It can be concluded that we are still at a stage with an ‘absence of evidence’ (and not necessarily an ‘evidence of absence’) when it comes to the translation of AI into clinically relevant impact, at least in terms of drug approvals and the delivery of therapeutic breakthroughs rooted in novel biology and therapeutics truly discovered or explained by AI.

By contrast, progress with AI-based technological capabilities is much more encouraging¹⁴. For instance, we are able to predict a variety of protein structures¹⁵, and design proteins with certain functions^{16,17}. We also have a huge swathe of novel measurement technologies to generate data that could be analysed with AI tools at our disposal, such as single-cell and spatial omics data¹⁸, as well as a greater range of therapeutic modalities. Although more capabilities and more data lead to improved predictive approaches when examined in isolation¹⁹, we have not yet seen evidence for better decision making when it comes to overall translation in the clinic.

Drug discovery and development involves a chain of decisions and pivotal points, which makes it difficult to identify the precise reasons why a project is successful (or not). Was it the selection of the right target, or was it due to using a more physiological or disease-relevant assay for primary screening? Was it luck that the right compound was in the screening deck, or chemists having the right intuition during lead optimization, or the availability of the appropriate translational animal model? Yet there is the tendency to ascribe failure to ‘bad luck’, but success to the method used, whether computational or experimental. Such attributional tendencies often feed into communication biases (similar to attributing R&D success to a ‘target-based’ compared with ‘phenotypic’ discovery²⁰) and can lead to biases in public perception. Therefore, it is difficult to assess the impact of AI in a particular project with certainty (as has been voiced for computer-aided drug discovery before²¹). This further confounds the analysis of whether AI is translating to improved clinical outcomes across the industry, or at least a subset of it.

In this Perspective, we first consider where the application of AI tools could be most valuable in drug R&D. We then discuss the characteristics and challenges of the data currently available to train these tools, as well as issues with model validation, highlighting a mismatch between the available data and the AI application areas of greatest potential value. After overviewing the current status of AI applications in various areas of drug R&D, we then provide recommendations on where to go from here to increase their translational relevance, as well as discussing non-technological factors preventing progress.

Where would improvements matter most?

Across the multiple stages of drug discovery and development, what potential improvements due to the application of AI tools could have the greatest impact? This is what we modelled in recent work²² and with updated numbers presented here, considering the impact of being faster, cheaper or better (that is, improving success rates) at each stage on the total capitalized cost of bringing one drug to successfully to market.

Clinical impact is what matters most (Fig. 1a) – specifically, higher success rates in the first phase of efficacy evaluation (usually phase II), consistent with the findings of a previous study²³. This phase has low success rates because it involves shifting the focus from the evaluation of safety, tolerability and pharmacokinetics in phase I to efficacy and optimal dosing in a larger patient group. In addition, outside of oncology, phase I usually involves recruiting a small number of healthy volunteers, making variability lower, whereas in phase II, diverse patients with different genetic backgrounds, disease severities and co-morbidities are involved, increasing variability and potential confounding factors. Phase II trials are also typically longer than phase I trials, and overall require substantial investment that represents a major contribution to the total cost of successfully bringing one drug to market.

However, current efforts using AI in small-molecule drug discovery largely focus on the preclinical phases, such as hit discovery²⁴. This focus is understandable, given that success in phase II is determined by the right selections made beforehand, both about the physicochemical characteristics of the drug candidate (even reaching as far back in the process as screening library choice) and the patient population in which to test the drug. Furthermore, preclinical costs and time taken do matter, both for individual organizations and in models of the total cost of bringing one drug to market. Hit discovery and optimization may need to be performed multiple times by an organization to lead to a single drug candidate, and many more times when considering the number of candidates that need to enter clinical development overall to eventually result in one successful drug launch, given the high failure rates. Nevertheless, the risk of applying AI only to the preclinical phases of small-molecule drug discovery (and in the ML literature, often to ‘ligand discovery’ rather than ‘drug discovery’; Box 1) is the analogy to a situation where one ‘looks for the keys where the light is’ – in this case, where the labelled data are that we can more easily model. This contrasts with looking for the keys where they are most likely to be located, which may not be where the ‘light’ (the data and feasible AI modelling approaches) is.

Empirically, general approaches aimed at increasing clinical success rates have been formulated, such as AstraZeneca’s 5R framework²⁵ and Pfizer’s ‘learnings on R&D productivity’²⁶. A focus on the understanding of disease biology is common across these frameworks, and also organizational culture – such as an emphasis on data-driven decision making, which ML often facilitates. The question is now how to support decision making in projects with data and AI tools that matter

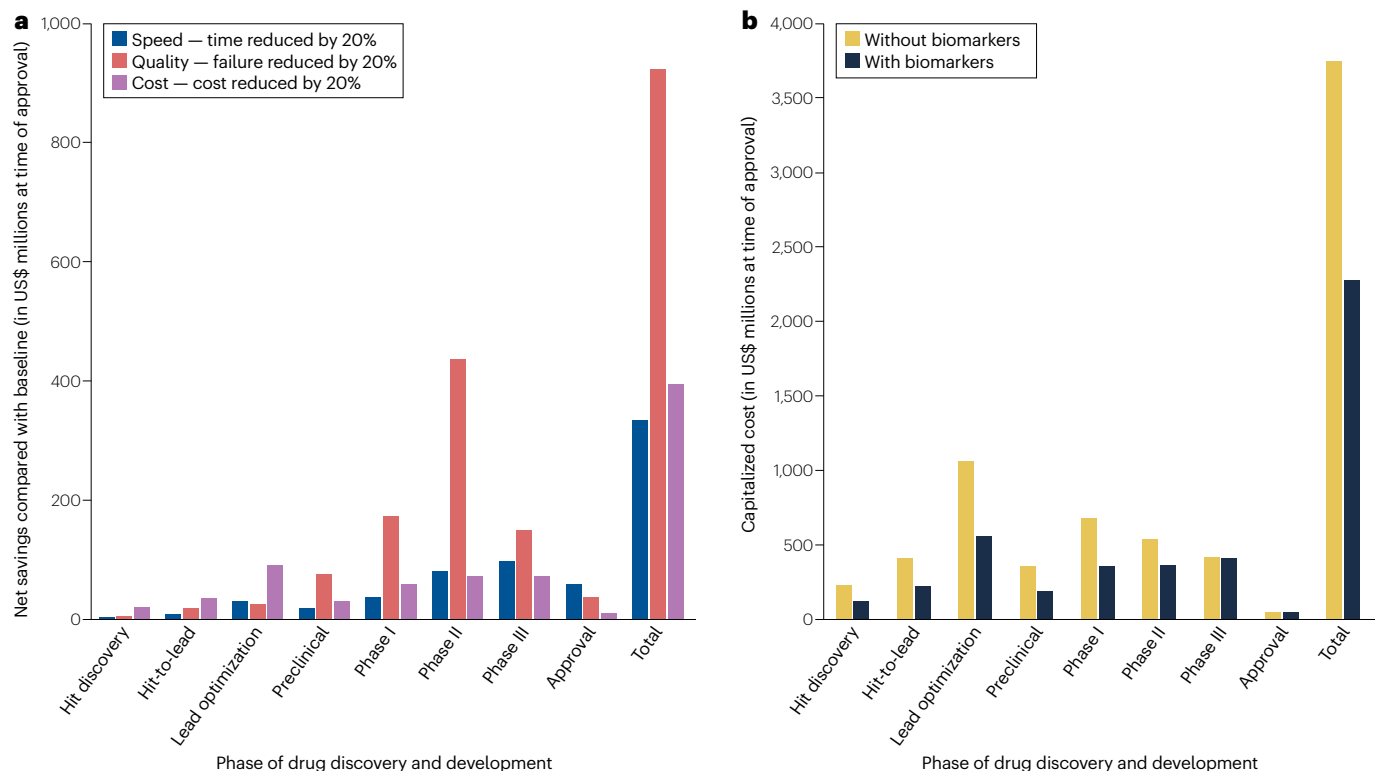


Fig. 1 | Impact of selected drug development parameters on capitalized cost per successful drug launch. **a**, The estimated net savings from increasing speed, reducing failure rates and lowering the cost of different stages of the drug discovery and development process (by 20% each, in relative terms in case of failure rate) compared with a baseline cost for one successful drug launch²². Costs per phase are based on ref. 23, with capital costs, as in this previous work, set at 11% and all costs capitalized and accumulated per successful launch. Shown are net savings per successful launch, compared with baseline assumptions. Reducing failure rates at phase II had the greatest impact. **b**, The estimated capitalized costs per drug launch of projects using biomarkers to better match drugs to the right

patients, compared with not using biomarkers²⁷. The cost per successful launch for projects using biomarkers in the dataset studied (~21,000 compounds in commercial databases of drug candidates from 2000–2015) was estimated to be approximately half that for projects that did not. The cost per trial is assumed to be constant in the model, and so this reduction in cost is due to increased success rates in the clinical phases and the resulting effect on capital expenditure in the preceding phases. Panel **a** is a corrected calculation of the one shown in ref. 22 and panel **b** is derived from a new calculation based on data provided in ref. 27. Input assumptions and calculations are available in Supplementary Table 1 and as a Python notebook in the Supplementary information.

for clinical impact. A key example of clinical impact using data for decision making is shown in Fig. 1b, comparing the cost per successful drug launch of projects using biomarkers for patient stratification compared with projects not using them (data from ref. 27). The capitalized cost per launch of projects that use biomarkers to identify the right patient cohort for a given intervention is only slightly more than half that of projects that do not use biomarkers for this purpose. Given that the cost per trial is assumed to be constant in this model, the halving in cost is solely due to increased success rates in the clinical phases (and the resulting impact on capital expenditure in the preceding phases). In practice, of course, cohort sizes also change, leading to an even more pronounced reduction of capitalized cost per launch. This is a clear demonstration of what meaningful impact in drug discovery looks like: tangible results (also) at the clinical stage, such as when genetic support for a target translates into higher clinical success rates²⁸, rather than impact limited to hit discovery and preclinical development.

Therefore, to increase the success rate of clinical trials and discovery of new drugs, the focus of AI in drug discovery must shift from doing what can be done – such as modelling data that is readily available, but that is unlikely to move the needle – to doing what should be

done, even if this requires, for example, substantial data generation or even data policy changes. Changes on this scale naturally move slower than new model-based iterations of AI algorithms, and hence could be a reason why we are yet to see clinical impact. It is more important to do the important things, than to do less meaningful things well. Or, in the words of John Tukey “Far better an approximate answer to the right question... than an exact answer to the wrong question”.

So which decisions, based on which parameters, can predictive AI models for end points in various stages in drug discovery projects support? An incomplete list of end points for which models have been established, that have various degrees of translatability to the clinic, is provided in Table 1, categorized by matching (1) the right drug, (2) dosed in the right way, to (3) the right patient, with a primary focus on small-molecule drugs. The models listed, both with respect to the particular end point predicted, as well as the particular implementation of the model from the data, representation and ML side, have very different relationships to translation to the clinic, and to efficacy and safety considerations. In many cases, compound parameters are measured, and fixed, early in projects, whereas the impact of such choices only becomes apparent in later clinical stages. For example, the logD

Box 1| The differences between ligands and drugs

Much as Goldstein and Brown reminded us in 1997²¹⁹, referencing Rene Magritte, “A gene sequence is not a drug”, a ligand, too, is not a drug. Press releases and articles often conflate ligand discovery and drug discovery, presumably because the term ‘potential drug’ sounds more attractive to investors than ‘ligand’. Such misuse of terminology understates the complexity of drug discovery — its translational aspects in particular — and makes it more difficult to understand which claims of a report are appropriately supported by the data.

The difference between a ligand and a drug is profound²²⁰ (see the figure and Supplementary Table 1). A common example of the difference is that a potent ligand might be toxic in vivo, and hence useless as a drug. A typical pitfall in this regard is affinity driven by lipophilicity, which often makes a compound unsuitable to be used in vivo due to its associated pharmacokinetic (PK) properties, such as its solubility, bioavailability and metabolism.

The test systems and the end point(s) measured for ligands and drugs are different. A ligand binds to a protein (or another target

of interest) in, at least, a biochemical system or assay, and end points might be limited to parameters such as half-maximal inhibitory concentration (IC_{50}), inhibition and dissociation constants (K_i , K_d). A drug is expected to bind to and engage its target and additionally demonstrate acceptable efficacy and safety as assessed by a range of preclinical and clinical end points.

These differences mean it is much easier to find a new ligand than a drug. The greater difficulty with identifying drugs compared to ligands is apparent in the number of known ligands compared with known drugs: there are $>10^6$ bioactive ligands in databases such as ChEMBL and PubChem but only on the order of 10^3 drugs on the market. In summary, the terms ‘ligand’ and ‘drug’ should not be conflated — and there is only a low likelihood that the discovery of a ligand will also lead to the discovery of a drug. ADME, absorption, distribution, metabolism and excretion.

Property	Relevance for efficacy and safety and relationship to other compound properties	Consideration in ligand design	Consideration in drug discovery
On-target activity	For ligand design, on-target activity is the primary, and often only, compound property considered by default	Sometimes only on-target binding is considered is ligand design, which does not guarantee necessary functional activity	Functional activity needs to be in a suitable range for a compound to become a drug
Physicochemical properties	Contributing to various target binding (for example, lipophilicity drives affinity) and ADME properties (for example, molecular mass and hydrogen characteristics influence blood–brain barrier permeation, lipophilicity is related to clearance, solubility needs to be achieved for absorption in the gastrointestinal tract)	Not by default considered if only on-target activity is the primary design objective	Physicochemical properties need to be compatible with on-target activity, later dose in man and dosing regime, and ADME/PK properties
Off-target activity/selectivity	On-target and off-target activity needs to be compatible with compound dosing to achieve desired safety and efficacy profile	Not by default considered if only on-target activity is the primary design objective	Off-target selectivity needs to be sufficient to achieve in vivo safety
In vitro ADME/PK	In vitro ADME/PK allows for some extrapolation to in vivo pharmacokinetics, and hence efficacy and safety estimation	Not by default considered if only on-target activity is the primary design objective	In vitro ADME/PK, although only a proxy measure, needs to be suitable for compound progression into later stages
In vivo ADME/PK (often multiple species)	In vitro ADME/PK in combination with on-target and off-target characterization of a compound allows for best efficacy and safety estimation	Beyond the scope of ligand design	In vivo ADME/PK needs to be compatible with on-target activity and dosing regime in humans

(lipophilicity) of a drug candidate is determined during its preclinical optimization, but whether sufficient bioavailability (which is heavily dependent on logD) is actually obtained only becomes apparent in phase I and later trials. This disconnect between early-stage decisions (where AI models can be readily used based on proxy data) and their impact in the clinic (which becomes apparent often years later) is one of the fundamental challenges in drug discovery, as well as in quantifying the impact of AI on this process.

Challenges posed by biological data

Biological data types relevant to drug discovery, such as data from cellular or in vivo disease models, pharmacological data and clinical trial data, are often fundamentally different from data in other fields in which AI has been successfully applied, such as image classification and

speech recognition²⁹. Although data in fields other than drug discovery may be scarce and high dimensional (and thus riddled with confounders) too, biological data are also highly conditional (depending on factors such as experimental conditions, genetic variability or environmental influences). Furthermore, we generally do not understand the underlying processes connecting input and output in a given experiment on a solid theoretical level — a situation referred to as epistemic opacity. The consequences for AI in drug discovery are profound: specifically, we often cannot assign meaningful and unconditional categories to, for example, compound effects, which prevents us from training AI models in a consistent and reliable manner to enable clinically relevant decision making (Table 2). The importance of these two aspects of biological data may be less widely appreciated, and so we expand on them here.

The highly conditional nature of biological data²⁹ means that, depending on a large number (dozens to thousands) of variables, the result of a biological perturbation will differ, even if only a single variable changes. For example, if the BCR–ABL inhibitor imatinib is administered to a patient with Philadelphia chromosome-positive acute lymphoblastic leukaemia, they will probably respond to the treatment – whereas other patients with acute lymphoblastic leukaemia whose cancer is driven by different aberrations will not. Similarly, some patients with cancer will respond to certain drugs, at a given and constant dose, whereas high expression of cytochrome P450 (CYP) enzymes that metabolize these drugs or low expression of CYP enzymes that activate prodrugs³⁰ in other patients might render the same dose ineffective³¹. Lifestyle choices can also have an impact. For example, smoking induces the expression of CYP isoforms³², affecting drugs that are metabolized

by those enzymes, and nutrition changes the gut microbiome, which can further alter drug response³³. Whether a given patient with a particular cancer responds to a particular drug depends on all variables of the system. Furthermore, even data gathered from streamlined biological systems under controlled conditions, such as cellular pharmacogenomic screens³⁴, can be remarkably inconsistent empirically.

The experimental parameters in data generation that influence the potential for successful data analysis run deeper than often appreciated by scientists focusing solely on data analysis. For example, there are multiple relevant questions when obtaining clinical samples for generating and analysing spatial and single-cell-omics data from patients with cancer that are not readily apparent from the data itself. Was it possible to obtain a sufficient number of samples from patients? Are the samples of sufficient size and quality? Are data on the clinical and treatment

Table 1 | Examples of clinically relevant end points that can be modelled computationally

Properties	How computational methods can help	Feasibility/required for clinical translation/challenges	Phase the parameter will be determined	Phase the parameter will show impact	Necessary experimental support
The right drug					
Physicochemical properties	Predictive models of various end points ¹⁹	Selection of compound properties related to both compound intrinsic activity and PK	Hit-to-lead/lead optimization	Preclinical to clinical	Assays
Target selection	Identification of disease-modifying targets (e.g., from GWAS or Mendelian randomization ¹⁴⁴ studies)	Drug targets are ~2.6 times more likely to be successful in the clinic if supported by genetic information than not ^{28,144} . Problems exist in case of multifactorial diseases, and in translating disease-associated targets to those which can be targeted with a given modality	Target discovery/early-stage fundamental research	Phase II (efficacy)/phase III (gold standard)	Early first-in-human studies to support target choice
On-target activity	Bioactivity ^{93,145} / off-target predictions ^{146,147}	Feasibility depends on target and assay end point; related to compound PK and dose in human model, resulting in the therapeutic index. Also impacted by pharmacogenomic variables (e.g., fast metabolizers)	Hit-to-lead/lead optimization	Phase II (efficacy)	Disease-relevant assays; target-based, and those showing target engagement in vivo
Off-target activity			Hit-to-lead/lead optimization	Phase I (safety)	Safety profiling assays
Metabolism	Metabolite prediction ¹⁴⁸	Absolute rates of metabolism are desirable (but more difficult to predict than relative rates); problem of combinatorial explosion	Early on using in vitro and in silico methods; later in vivo	Phases I–III (safety/efficacy)	In vitro and/or in vivo assay systems
Dosed in the right way					
In vitro-to-in vivo extrapolation	Estimating, e.g., dose in humans from in vitro data ¹⁴⁹	Sufficient knowledge about mechanisms and selection of compound properties for in vivo extrapolation	In vitro phases	Preclinical and clinical phases	In vitro/animal/human PK studies
In vivo PK	Direct in vivo PK prediction, in animals ^{67,150–152} and in humans ^{58,153}	Coverage of chemical space with in vivo PK data is often not sufficient; wide variance between individuals and studies; but fast and often useful to remove, e.g., compounds with too high clearance at design stage	Early on using in vitro and in silico methods; later in vivo	Phases I–III	Animal and human PK studies
Matched to the right patient					
Patient endotyping	Bioinformatics methods can help identify biomarkers and define patient cohorts ^{27–29}	Sufficient understanding of disease endotypes and patient cohorts required to predict treatment response (can be related to mode of action or, e.g., PK aspects of treatment)	Can be very early (during target selection) or later (during clinical trials)	Phase II/III	Sufficient clinically relevant data to allow for endotyping
Patient stratification ^{27–29,139}	Using predictive biomarkers to distinguish likely responders and non-responders	Ability to identify predictive biomarkers is often hampered by the unavailability of suitable datasets, and/or insufficient disease understanding	Can be very early (during target selection) or later (during clinical trials)	Phase II/III	Sufficient clinically relevant data to allow for patient stratification

The end points in the table are primarily related to small-molecule drug discovery. It can be seen that many parameters in a drug discovery project are determined early, with the consequences of those choices only becoming apparent in the clinical phases. The need for this translation must be taken into account for early in vitro end points and, in the context of this article, also for any computational approaches applying artificial intelligence to support drug discovery²⁹. GWAS, genome-wide association study; PK, pharmacokinetics.

Table 2 | Labelling challenges for data from preclinical and clinical experimental set ups

Aspect of experimental set up	Preclinical factors that prevent clear label assignment	Resulting problem in preclinical context	Clinical factors that prevent clear label assignment	Resulting problem in clinical context
Compound structure	Conformations, protonation states (for example, pH in GI tract), tautomers, metabolism/reactivity	Compounds exist in multiple states (and it is not clear which one to use for computational models in all cases)	Crystal polymorphs, salt forms	Even during GMP production, polymorphs can change, with resulting impact on solubility and hence bioavailability ¹⁵⁴
Compound administration	Solubility and permeability limitations	Often in vitro assay results are confounded by insufficient solubility, or other compound properties leading to assay interference ¹⁵⁵	Pharmacokinetics/target engagement	Response to treatment is dependent on target engagement, which in particular for CNS targets is not trivial to establish
Set up of biological test system	Variance due to cell line, dose, timepoint; characterization of biological test system	Biological test systems are always incomplete proxies of the in vivo situation and not trivial to characterize ²⁹ ; several (often arbitrary) assay parameters influence readouts	Biological test system (human and disease state) are always incompletely characterized	Clinical trials are controlled and randomized, but they still often do not sufficiently characterize the system (human) to understand response to treatment (related to both efficacy and adverse drug reactions)
End point measured in test system	Measurement depends heavily on experimental set up (for example, ATP concentration in kinase inhibition assays)	Data between assays can often not be combined. Also, isolated proteins behave differently from in vivo systems ¹⁵⁶ ; for example, in terms of downstream signalling or receptor dimerization	Clinical success depends on end point definition	Clinical end point can be chosen arbitrarily (such as survival, tumour size or surrogate end points) and compared in different ways (such as versus standard of care or placebo), hence clinical success depends on end point definition
Performance metrics used	Diverse performance metrics used, including precision, recall and F1 score	Metrics used for model evaluation are not linked to use case ⁵⁵ (see also the main text)	Diversity of response across patient cohort	Heterogeneity of responses across patient cohort, difficulty to summarize in a single number and to generalize

The conditions used to generate data types in the table are very difficult, and in many cases impossible, to account for in predictive machine learning models, as either one creates ‘clean’ data subsets that are too small, or one uses larger, but heterogeneous, datasets for model training. It may be tempting to let the algorithm infer the missing parts, which is sometimes possible, but by and large, in particular in more complex experimental set ups, this is not possible due to the high conditionality of the labels on a very long list of such assay parameters (see also ref. 29). CNS, central nervous system; GI, gastrointestinal tract; GMP, good manufacturing practice.

history available and accurate? Can we match with suitable controls for analysis? Data might appear ‘clean’ once available for analysis, yet its appearance often belies a difficult and messy journey to get to this stage.

On the side of output and ‘predicted’ variables, we encounter the same problem – output variables, such as the response to treatment of an organism, or cellular response to compound treatments, depend on the assay end point selected. Is a compound ‘active against a protein target?’ This depends on the assay used, whether it is a binding assay or functional³⁵, and on the assay parameters themselves (for example, ATP concentration in kinase assays). Is a compound ‘efficacious against a disease?’ This generally depends on the particular endotype of the disease, how the compound is dosed, and the choice of clinical or other end points³⁶. More complex biological end points are difficult to describe with labels, but these are required in supervised AI models that are meant to contribute to decision making.

In practice, data can often be at best slightly useful, but at worst misleading, when possessing insufficient predictivity validity²⁰ for the actual outcome of interest. Two examples of this are shown in Fig. 2 to illustrate how data in drug discovery often looks like in reality and the practical implications. The aim of the application illustrated in Fig. 2a is to predict drug-induced liver injury liability using data from screening compounds in liver organoids, but the correlation between clinical drug-induced liver injury risk in patients and the in vitro assay readout is low, even when the readout is corrected for human plasma exposure (C_{max}) – it would be even closer to zero otherwise. In the application illustrated in Fig. 2b, thermal shift – a measure of compound binding affinity to a target that is amenable to high-throughput measurement – is only marginally associated with harder-to-measure IC_{50} activity against the target being studied, hence also not allowing for conclusive prioritization. Furthermore, these results are context specific;

doing similar assays with marginally different methods would give quantitatively different results. Solely feeding such data into AI models would not result in better individual decisions, and hence not better clinical outcomes. We argue that evaluating the relationship between data and useful outcomes should be mandatory, both before (using the input data) and after (using the predictions) model development. In practice, however, researchers are often constrained by the data that happen to be available, which are therefore used, and sometimes misused, in AI applications.

The highly conditional nature of so much of the available biological data in drug discovery also complicates the application of AI, as many ML approaches require labelled data. This is true for supervised approaches, but also generally, to make decisions in projects, decision boundaries are always needed. Yet how can we appropriately label, given that the available data are often conditional, inconsistent or of varying quality? Furthermore, how do we capture the impact of all the relevant experimental properties in a model if such properties are unknown or unmeasured? Any ML method in drug discovery that does not consider these aspects of data will have severe limitations when it comes to the relevance of its predictions.

For higher-level readouts such as in vivo readouts, there is a systematic problem assigning labels, in that there is no agreement regarding which labels matter and what constitutes a label. For example, historically used labels, such as International Classification of Diseases (ICD) codes for medical billing purposes, are now used for a completely different purpose, such as output variables for AI models. To illustrate this in more detail, disease ontologies aim to classify diseases. However, given that the underlying disease pathways are insufficiently understood, disease classification is currently often descriptive and on the organ level, instead of on a level that captures actual disease biology³⁷.

For example, depression, anxiety and anorexia are different ‘diseases’ according to classifications such as ICD-11 or the Experimental Factor Ontology (EFO) – but symptoms (and, in fact, underlying genetics³⁸) are shared, so how should one train a ML model on such labels? Using both disease labels as distinct does not do the shared symptoms justice. This is further complicated by the fact that symptoms may be shared often only in some patients (but not others), some of the time (but not constantly) and to different degrees. All of those aspects make just ‘labelling data’ currently very difficult, in particular if one goes further to the in vivo side of disease, and treatment effects (and also adverse drug reactions). Our understanding of disease needs to be anchored in biology, and not only symptoms.

Challenges posed by chemical data

A major problem with data in the chemical domain used in drug discovery is that chemical space is large and high dimensional, and data in practically available datasets insufficiently samples this space and is biased. Here, two aspects are particularly important: the implications of the size and fragmentation of chemical space for model validation, and the consequences of biased datasets for the current trend of ‘explainable AI’.

The problem of generating and validating a predictive model in chemical space is shown in Fig. 3a, with the universe used as an analogy to chemical space, and where every galaxy represents an area of chemical space, with stars and solar systems representing a more

densely defined space. This is a common analogy, as the observable universe is thought to contain (at least) 200 billion galaxies and around 10^{24} stars. In comparison, chemical space is probably even larger – up to $\sim 10^{60}$ compounds in small-molecule space³⁹ – and for all practical purposes infinite when it comes to macromolecules and assemblies thereof. Practical datasets can therefore comprise only a small fraction of this space. Furthermore, chemical space is very local in nature – appending a particular functional group to one scaffold can have very different effects than appending it to another molecule. This makes both sampling and extrapolation in chemical space fundamentally difficult⁴⁰ and practically only possible when sufficient training data is available, descriptors capture underlying trends in properties and the directionality of features plays a negligible role – such as in the case of logD or other related physicochemical properties¹⁹. Even working with ‘solubility’ is much more difficult – namely due to the solid state structure of small molecules, which is intrinsically based on directional interactions between molecules, but which is not apparent from the structure of a single molecule itself.

The inherent local and biased nature of models poses systematic limitations on what small-molecule predictive models will be able to achieve in the future when it comes to modelling global property space in a single model. Validating models based on time-series splits⁴¹ aims to account for these limitations⁴²; however, it raises the question of how closely this validation scenario reflects true future applications, which will often involve entirely different ‘chemical series galaxies’ (Fig. 3a).

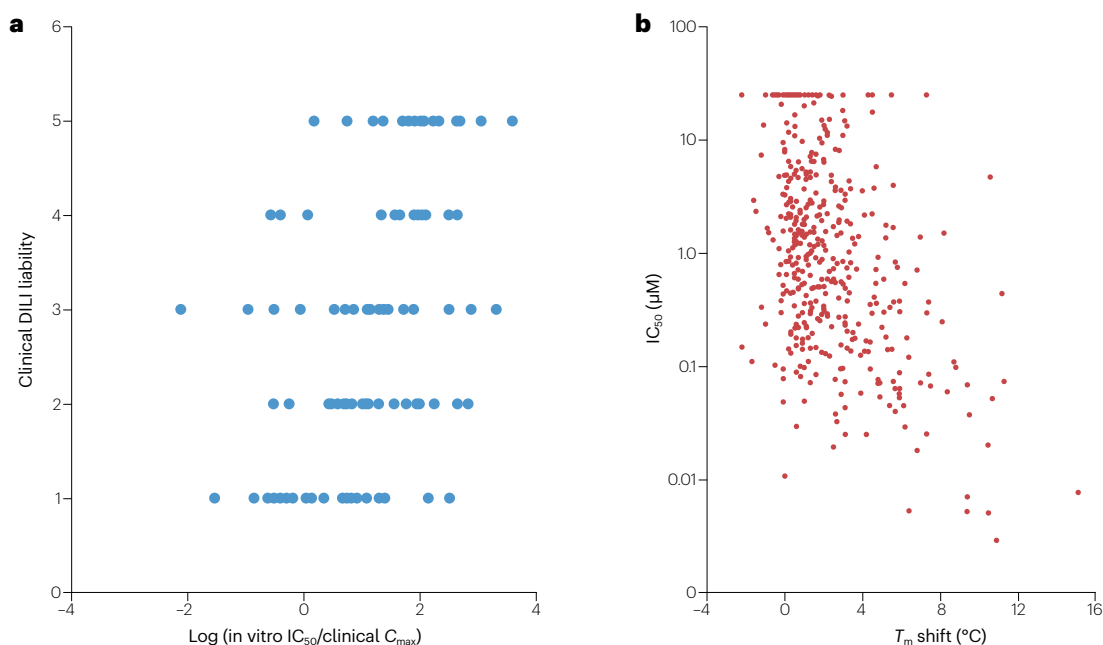
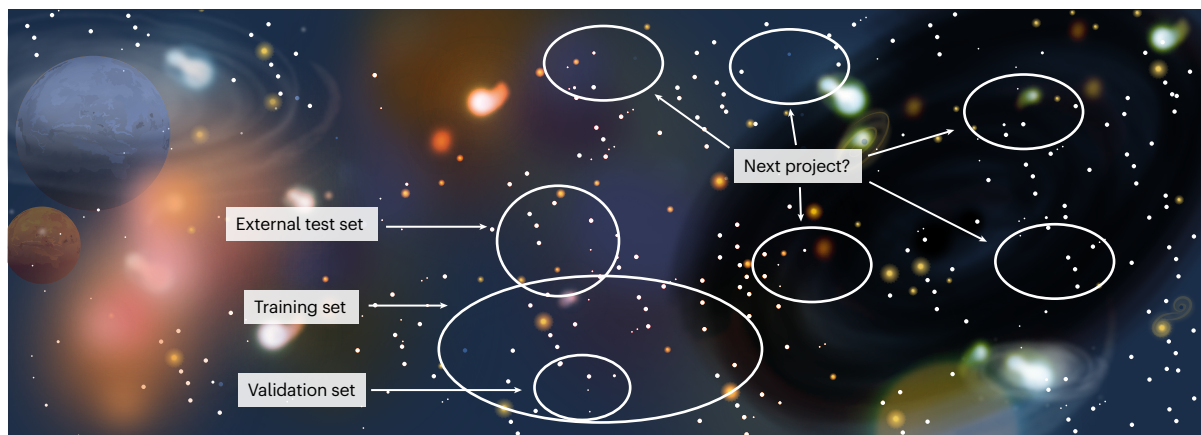


Fig. 2 | Illustration of the low predictivity of proxy data currently available in drug discovery. **a**, The graph illustrates the relationship between half-maximal inhibitory concentration (IC_{50}) values for drug-induced cytotoxicity derived from liver organoids (corrected for human plasma exposure (C_{max})) and clinical liability for drug-induced liver injury (DILI) for a panel of drugs with known levels of DILI liability, showing the low correlation between both values (lower liability index values indicate higher clinical liability). There is a large overlap between clinical risk classes despite the complexity of the liver spheroid model and the correction for human drug exposure, and hence the predictive validity of this readout is low. Data are from Supplementary Table 3 in ref. 216. **b**, The graph

illustrates the relationship between thermal shift assay data (in which binding of a compound leads to stabilization of the protein, observed as a shift in its melting temperature (T_m), x axis) and enzymatic inhibition (IC_{50}) data, y axis). Although it is easy to generate, the thermal shift data have relatively low predictivity for enzymatic inhibition. The graph is adapted with permission from ref. 217, CC BY 4.0. These are typical cases of data used in the drug discovery context, and given the low concordance of the proxy metrics with the biological measure of interest (here DILI liability and enzymatic inhibition, respectively), feeding available data into ‘artificial intelligence’ models will not lead to better individual decisions, and hence clinical outcomes.

a



b

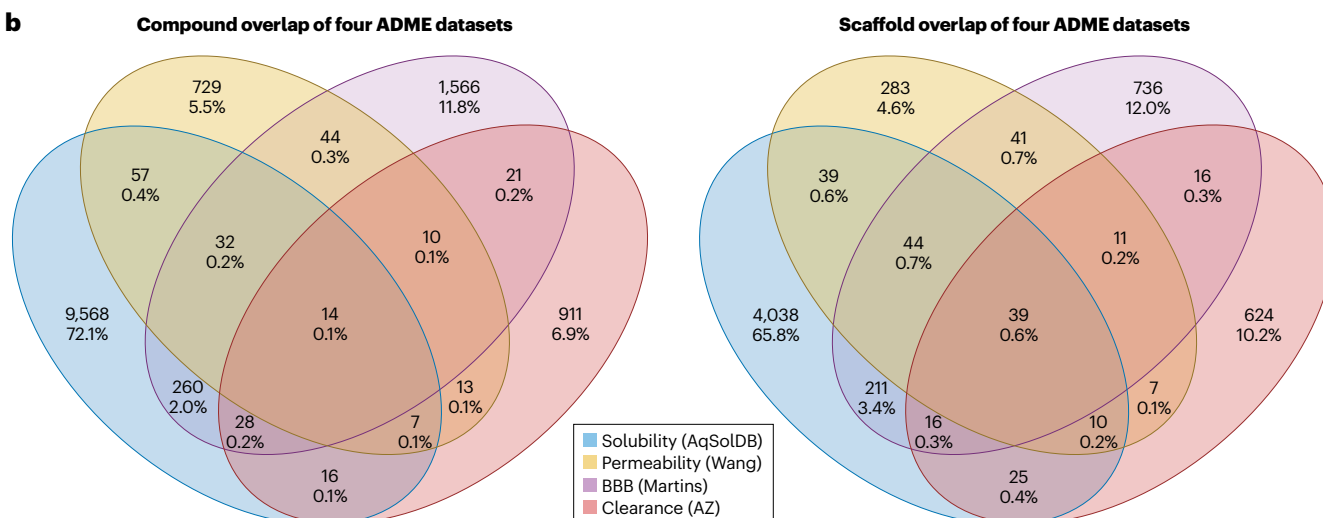


Fig. 3 | The challenges of chemical data in predictive modelling.

a, An illustration of the nature of chemical space (to the extent this can be visualized in two dimensions). Given the location of a new project in chemical space is, by definition, unknown at the timepoint of model generation and performance estimation, performance on a test set does not extrapolate to future project use cases. In addition, so-called 'external' test sets are often not truly external, but rather a subset of data available at the time a model is generated, and hence they tend to overestimate model performance in comparison to future project use in regions of chemical space beyond this. Adapted from ref. 66, CC BY 4.0.

b, An illustration of the structural overlap between four datasets related to absorption, distribution, metabolism and excretion (ADME) from the Therapeutics Data Commons²¹⁸ (for water solubility, membrane and blood–brain-barrier (BBB) permeability and clearance) that are commonly used for machine learning model generation and benchmarking, based on canonicalized SMILES notations (left) and Bemis–Murcko scaffolds (right). The relatively small size of such datasets for machine learning and the very small percentage overlap of data between datasets are apparent, leading to different applicability domains for different models. This is despite the need to model all properties for a specified chemical space.

Current methods for defining applicability domains in chemical modelling are similarly constrained, as they cannot anticipate future use cases that are, by definition, unknown.

It is also crucial to remember that therapeutic design is a multi-objective optimization problem, requiring an optimal profile balancing all the properties outlined in Box 1 for small-molecule drug candidates. Hence, this problem of local and biased chemical spaces with limited known predictivity for future projects is further exacerbated by chemical space compatibility between models trained on different chemical spaces to predict different end points. A simple illustration of this problem is shown in Fig. 3b, where four common ADME-related datasets for different but necessary end points share little to no overlap

in training data and hence the resulting applicability domain. Only 0.1% of compounds and 0.6% of scaffolds are shared among all datasets. This represents a notable shortcoming, particularly when only open-source datasets can be used in harmonizing ML models of molecular properties across different end points. Without synchronized applicability domains, multi-objective optimization becomes difficult and risks over-reliance on model generalization.

These fundamental problems also contribute to the large disconnect between publications that report promising performance values and the subsequent performance of the model when applied to a future project. In publications, very often no prospective validations are performed, and then test sets closer to the training set (due to being

split from the same original dataset) effectively lead to ‘information leakage’⁴³ and an artificial increase of performance on the test set, which does not translate to future projects. There are also psychological effects involved: it is difficult to convince scientists to put in extra work that will make their results look worse, especially when some ML specialists entering the field lack the requisite domain knowledge and prefer to spend time on the model architecture rather than the data and its meaning in a given context. A prototypical way of thinking from the side of an experimental scientist and a computer scientist is visualized in Supplementary Fig. 1, clearly illustrating the dangers of lack of communication and understanding of each other’s domain.

Explainable models (XAI) can also lead to pitfalls in this field due to the inherent bias in data used to train models. An example of this is illustrated in Supplementary Fig. 2. In this example⁴⁴, the goal was to identify chemical features associated with bitterness across a dataset of several hundred bitter molecules. Using an information gain criterion on a fingerprint representation, the features flagged as correlated with bitterness were, rather surprisingly, sugar rings (and parts thereof). Although some sugars are known to activate bitter receptors⁴⁵, as a general finding, sugar rings are of course very unlikely to be causally related to bitterness as the model suggests. They rather represent a spurious correlation in the data caused by dataset bias, as natural products are frequently glycosylated⁴⁶ and natural products also frequently have bitter taste. Although in this case the spurious correlation might be obvious, this is not necessarily the case in other applications of XAI methods applied to chemical datasets. In particular for datasets that were not created by experimental design for training ML models, but rather using project data, large series biases exist due to the nature of this dataset, leading to very similar biases as in the above example. If one looks into databases of any type, such biases (synthesis bias, confirmation bias, project bias, reporting bias and so on) abound, making the fruitful application of XAI algorithms in this area challenging. It should be emphasized that this is not a shortcoming of the algorithms used, which are often powerful, if applied on the right data. The issue is that chemical datasets, in the way they are available in real-world situations, do not belong to this category of data owing to data sparseness and sampling biases.

Generating data for AI applications

Data are not knowledge – understanding a process functionally and parametrizing it with data to support real-world decision making, is. For drug discovery, the number of variables involved in both living systems and experimental set ups is so large that data alone are insufficient to answer key scientific questions. Furthermore, data are frequently collected in the expectation of a benefit that may not be tangible, and potentially even a downside in cases where the user is unable to establish where to find the signal in a large volume of data to answer the question of interest, in effect then just resembling noise.

Another key problem is that the nature of the data at hand in drug discovery – its confounding factors, conditionality and epistemic opacity – may not be understood by everyone aiming to model such data, and hence the limitations of labelling data and using labelled data are not understood either. As a consequence of ‘believing the labels’, there is a considerable knock-on effect on benchmarks that are being used in the field. Benchmarks have been crucial to provide some estimate of progress in AI, given that otherwise no quantitative comparison of novel approaches to the existing ‘state-of-the-art’ (SOTA) would be possible. In some areas, such as image recognition and speech recognition, benchmarks by and large translated to better

performance in real-world use cases. However, in the area of data used in drug discovery, this has to be viewed with caution, as the conditionality and often ambiguous nature of such data often does not lend itself easily to the construction of ML models²⁹. Hence, this represents an insufficiently specified problem setting⁴⁷, with the resultant gap between numerical improvements in benchmark situations and translation to improved decision making in real-world drug discovery projects. The focus on benchmarking datasets, even when considering best practices for ML in the field⁴⁸, can hence even be misleading and create an illusion of progress – ‘we are able to predict better on the benchmark dataset’. However, for translation to real-world decision making, this criterion is not sufficient (and arguably not even necessary). ‘SOTA-chasing’ is a term that has been coined for this tendency in some domains of ML⁴⁹. Goodhart’s law states that when a measure becomes a target, it ceases to be a good measure, and one manifestation is that available benchmarks can be gamed. In our view, benchmarking studies that do not aim to ‘sell’ a new method are more credible and a valuable addition to the literature if they use a relevant end point, are unbiased and not glamorous⁵⁰. Blind and prospective assessments such as the Critical Assessment of Techniques for Protein Structure Prediction (CASP)⁵¹, the Statistical Assessment of the Modelling of Proteins and Ligand⁵² or the Critical Assessment of Computational Hit-finding Experiments⁵³ are some of the few ways to assess performance accurately, but these are related to end points for which the conditionality of labels is usually not considered. In addition, they require substantial community effort and depend on the availability of prospective, labelled data to serve as a suitable ‘ground truth’ for evaluation, which is something that is not always readily available for many types of practically relevant problems.

We consider that generating data is not the bottleneck in drug discovery – it is ‘reduction to practice’, using it to inform clinically relevant decision making. One example of this is the Human Genome Project^{54,55} (and sequencing data more generally), which has certainly allowed us to create a (static) view of building blocks of biology, and which can be a key first step in understanding a biological system. However, some applications of sequencing data have not yet met early expectations. For example, for target identification and validation, the prediction⁵⁶ in 2001 that “The number of drug targets will increase by at least one order of magnitude and target validation will become a high-throughput process” has not yet been fulfilled.

Data have been identified as the fundamental problem in drug discovery before²⁹. However, often a focus on ‘quantity’, ‘quality’ and/or ‘better data’ is voiced. Although it is true that we need data that is sufficient in quantity and technically reproducible to properly separate relevant physiological responses from the noise of human and experimental variability, the main focus needs to be the predictivity of the data for in vivo outcomes^{20,29,57}. In other words, there is no point in generating data in an entirely reproducible manner if it does not sufficiently enable us to address relevant complex biological questions.

There has been lots of activity related to ‘data’ in drug discovery over the past two decades, and some of it indeed for the better. Companies have annotated their data repositories, so we now understand much better what we know, which is a great asset in itself, even without AI. Metadata annotations have generally improved, and cross-industry consortia such as MELLODDY⁵⁸ have aimed to make use of data from different companies to improve model performance. The question is how to progress from here – and there are several problems with this transition.

Compared with geography, where the map of the world is largely static and so can be used practically to move from A to B, biological data are heterogeneous between individuals, within an individual and over time, and it also changes as a function of many factors, such as ageing and nutrition. Furthermore, biological data are conditional²⁹, as noted above. Hence the analogy of building a ‘map’ of biology fundamentally does not hold. Furthermore, we do not even really know what ‘A’ and ‘B’ are in many cases – for example, when classifying disease or cellular states, it is often not unambiguously defined which markers (such as surface markers) one needs to look for when assigning disease and cellular states to a system.

Applying AI tools in the area of protein structure prediction – where we have labelled data – has been fruitful because we have a ground truth. For example, AlphaFold⁵⁹ falls into this area, as it is built on about 50 years of data deposited in the Protein data Bank (PDB), where electron density (with all its shortcomings⁶⁰) represents a suitable ground truth for model generation and evaluation. Such a scenario is not the case for many other desired predictions.

Model validation by itself is not sufficient

According to George Box’s famous maxim, “all models are wrong, but some are useful” (for a more recent revisit of the topic on clinical models, see ref. 61) – but which ones are useful, and for which precise purpose?

Despite the remarkable progress in biomedical science in recent years, developing safe and effective new therapies remains a difficult and risky business⁶², as evidenced by relatively constant, low rates of candidate molecules advancing to FDA approval. Where does this discrepancy between such promising technological advancement and such limited translational success come from? A critical, underappreciated factor may be mistaking ‘the map for the territory’, in the classic framing of Alfred Korzybski. Specifically, our tendency to overemphasize models and insufficiently focus on processes (where even processes are only an abstract approach to solving a complex problem⁶³).

Figure 4 depicts the integration of a model into the process of decision making based on earlier work⁶⁴. The model-based view, validation and quantitative assessment of model performance is solely limited to the inner, or ‘model’ box, and in virtually all cases is based on subsets of the training data used. In this case, the view is that ‘the model improves

in performance’ if numerical improvements on a certain subset of the data (for example, a ‘test set’) using a certain performance measure (for example, precision or recall) is observed. However, a model does not exist in a vacuum. It operates within two critical contexts: (1) the specific project it is part of (Fig. 4, left-hand side) and (2) the decisions its outputs are intended to inform (Fig. 4, right-hand side). Those two connections, to the project context as input and the follow-up as the output, are key for practical model application, as illustrated by the following example.

When training and validating a model – for example, for predicting toxicity at a given compound concentration – the concentration is a parameter set and chosen at the time of data generation, based on which a model will then be generated (Fig. 4, point (1)). The actual dose of a query compound that will be used later in humans (and thus its concentration in vivo), however, is unknown; such parameters are only known at the time of model use within a project (Fig. 4, point (2)). As a result, a model trained using one cut-off value for concentration may not be relevant if the concentration reached in humans differs. Even if the model can take concentration as an input parameter, its accuracy is limited by the range of concentrations represented in the training data. Let us now assume that the model is used anyway – what would be the real-world impact of that? Decisions in therapeutic programmes are not based solely on computational predictions. Therefore, prioritizing compounds using a predictive model poses the question of which follow-up experiments are rapid, cost-effective and translationally relevant to clinical outcomes (Fig. 4, point (3)). Without paying attention to these two interfaces (project context and contextual decision-making factors) – and instead just ‘validating the model’ alone – there is a loss of translation to a situation where the model is meant to support practical decision making. In short, models need to be linked to processes they are used in.

There are also aspects of model training and validation that require understanding of the process in order to derive a fit-for-purpose model (Box 2). For example, are we in a ‘discovery setting’, trying to identify some (hopefully diverse) active compounds in a very large chemical space? This is a selection setting where precision among the top-ranked (many) compounds matters most⁶⁵. Or are we filtering compounds⁶⁶, such as those with likely toxicity or very high clearance? This is a deselection setting, where the ability to avoid selecting compounds with

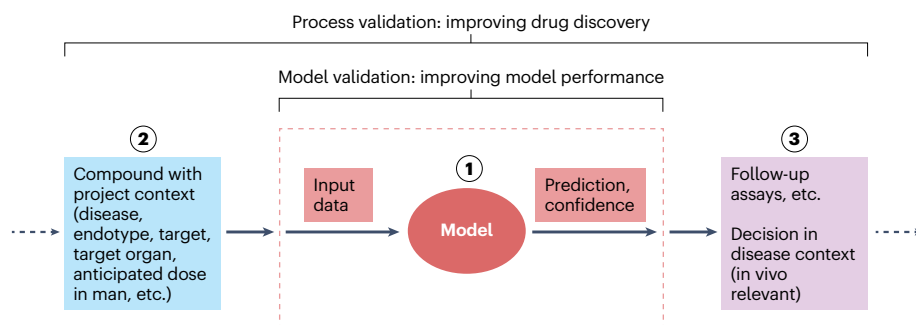


Fig. 4 | Visualization of the difference between model validation and translating a predictive model to improve an associated process; in this case, drug discovery. Model validation is shown within the box in the centre of the figure (1), and technical validation, as well as publications, often focus solely on this aspect, with the aim to improve upon a certain model metric. However, in real-world situations, models are used in the context of a process that provides crucial project context (2), as well as possibilities to follow up model

predictions (3), which need to be taken into account. Hence, for practically relevant model generation, we advocate moving on from model validation by itself and to consider its embedding into the use case from the onset. This then requires consideration of the choice of suitable data types and model construction choices for a given prediction purpose, how to learn from clinical observations to enable model updates and any other aspects needed to translate improvements in model performance to improvements in process performance.

Box 2 | Assessing performance of machine learning models based on the context of use

Machine learning models can contribute to different types of decision-making in drug discovery:

- In selection settings, the aim is to prioritize some (out of many) possible solutions to a problem, often at an early stage, such as during hit finding in a drug discovery project. Here, aspects such as ‘hit rate’ (or, in modelling terms, precision) will be relevant in a real-world scenario and can be used for meaningful benchmarking of models.
- In deselection settings, the aim is to identify solutions that are not good enough to achieve the overall goal; for example, to identify compounds for which clearance is too high, or which are likely to be toxic in vivo. These types of models can be applied at an early or late stage, and here the recall among the undesired class (the ability to screen out as many undesired molecules as possible) is often relevant in a real-world situation.
- In quantification settings, the aim is to predict a quantitative value for a given situation, such as the dose needed for a molecule in vivo to achieve the intended therapeutic effect. Here, the molecule is a fixed entity, and no prioritization among other compounds is desired — rather, the aim is to assign the molecule a quantitative value as accurately as possible. Here, measures such as the ‘root mean squared error’ of predictions (as a point-based error measure, as opposed to correlation coefficients, for example) can be a useful model performance metric.

The performance of a model therefore needs to be evaluated in the context of its eventual use case.

characteristics at the undesirable end of the range being predicted by the model matters most^{67,68}. Or are we trying to make a prediction at a later stage, for a compound that is to be dosed in animals or humans? This is neither a selection nor a deselection setting, but rather a situation where quantitative predictions for this compound, and a good understanding of the error involved, matter.

However, if computational models are only optimized with respect to a generic model aspect such as high area under the curve — a frequently used performance metric for the ability of a binary classifier ability to distinguish between classes across all thresholds — the quantitative performance evaluation will not be relevant for situations encountered in a real-world project. The problem of unsuitable metrics for model evaluation has been recognized in drug discovery^{65,69} as well as in other areas^{70–73}, and is illustrated in Fig. 5, which provides an example where approximately similar models with respect to overall performance based on area under the curve show very different behaviour in compound selection and deselection settings. Hence, model performance depends on the use case, and validating a model without taking the real-world use case into account is one of the contributing factors that prevents the translation of model performance into practical settings. In other words, use-case-based model metrics are key when evaluating models in a practically relevant manner.

A current direction to address this problem is to take property-specific model quality and decision-making thresholds into account both during model training and during visualizing results to the end user. For example, microsomal stability is a noisy measurement, and so an R^2 value will not provide a measurement of model quality that is relevant for practical purposes. Given that microsomal stability is only an early (and approximate) indicator of compound stability, a three-class or four-class categorical model would probably be more suitable here, with interval ranges chosen to correspond to real-world decision-making boundaries, hence rendering this model much closer to the use case. In a lead optimization setting, however, the main criteria would be the ability to pick out the highest-affinity ligands from the rest. Metrics such as enrichment factors, or more practically the hit rate within the subset of compounds that can be tested experimentally, are more meaningful in this context⁶⁵.

The routine reporting of generic performance measures in the literature helps explain why apparently strong published results often

translate poorly to real-world decision making. Improvements in numerical accuracy, even when genuine, rarely lead to proportional gains in downstream impact. In ML terms, this represents the problem of ‘underspecification’, which also has been recognized as a considerable hurdle for estimating model performance beyond the drug discovery area⁴⁷.

Furthermore, the models are often not the limiting factor, but rather the underlying data available (as highlighted above), and we have observed a tendency in applications of AI in drug discovery to ‘squeeze out’ marginal improvements in model performance when the data used for training often has considerable error^{74–76} that is neglected, despite the availability of methods to account for experimental uncertainty. The limited benefit of relying on marginal data when striving to maximize model performance, which hence has its upper limit, has been recognized before^{77,78}. Data quality in this context extends far beyond summary assay metrics such as z' score. Inherent confounders that arise from insufficient sample sizes or unaccounted biological batch effects such as cell death or inconsistent cell confluency can greatly influence model performance, and sometimes are not accounted for even in workflows with robust quality control metrics. Of course, the model (or modeller) cannot be held responsible for prediction errors close to the error of the underlying data — in fact, trying to be ‘better than the data’ is known to overfit models with possible numerical improvements on supposedly relevant test sets, but which will be detrimental to model performance in future projects nonetheless. This type of ‘overoptimism’ on model performance seems to already be present in some reported studies⁷⁸. A realistic view on data, use cases and modelling method used is key.

‘Prospective validation’ is intrinsically related to the evaluation of models on novel chemical and biological space, which in ML terms most often corresponds to out-of-distribution predictions. Given the vast dimensionality of chemical space, essentially all predictions fall into this category, making reliable performance estimation exceptionally challenging^{40,79} and leading to the frequently voiced observation that ‘every model is a local model’.

In summary, it is not sufficient to have a model that numerically performs well — we need to see it in the context of its use case, from every angle.

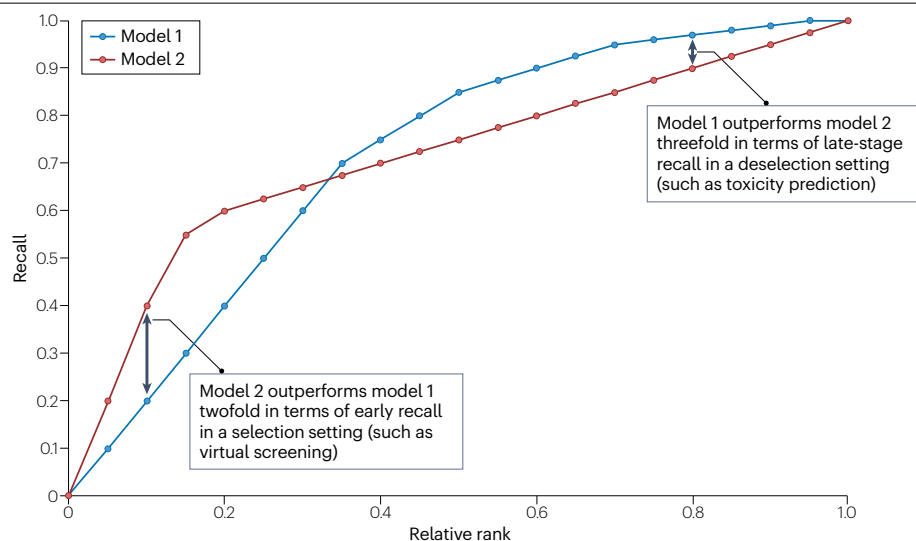


Fig. 5 | The importance of context of use when evaluating model performance. Two hypothetical models with approximately similar behaviour with respect to a frequently used model performance metric (area under the curve (AUC)), but very different behaviour in real-world use cases, are shown. The AUC here indicates the capability to identify the desired class of items (y axis, the higher the better), overall items in a prioritized list of items; for example, a subset of compounds to be experimentally tested out of a large screening library. Model 2 outperforms model 1 approximately twofold in an early selection setting. If 10% of the top-ranked compounds in the library are selected, the proportion of compounds that are actually in the desired class when tested experimentally would be 40% in case of model 2 compared with 20% in case of model 1. However, when applied

in a deselection setting, such as (late-stage) identification of toxic compounds, model 1 outperforms model 2–3-fold. If 80% of the top-ranked compounds in the library (in this case, ordered by their predicted risk of toxicity) are deselected and the remaining 20% tested experimentally, 3.3% of the compounds identified using model 1 would actually have undesirable characteristics when tested compared with 10% of compounds identified using model 2. These two situations illustrate the problems with generic model validation using the AUC as a performance metric without taking the use case into account, which makes it impossible to predict model fitness-for-purpose for a real-world situation⁶⁵. Differences in the model performance metrics used is one of the reasons why reported performance metrics and the performance of models in real-world situations often differ greatly.

Where are we?

A brief summary of some fields related to the use of AI in drug discovery is given in Table 3, to provide a snapshot of where the field currently stands. We also highlight two general application areas with notable successes in more depth here. We acknowledge that the SOTA in companies may be further advanced, but it is not possible to assess this yet without published examples as well as proof points of clinical translation.

Particularly impactful examples of AI applications include AlphaFold⁵⁹ and related generative AI protein folding methods such as OpenFold⁸⁰. Testament to this is the AlphaFold Protein Structure Database, which is reported to have had ‘over two million users in 190 countries’, all thanks to a collective database of ground truth protein structures deposited in the PDB over decades (not to forget the leveraging of millions of years of evolutionary fitness data). However, this is successful in the context of model validation – in this case, protein folding – and not necessarily process validation in a drug discovery setting, where the aim is to discover a modulator of such a protein with overall properties suitable to be progressed as a drug candidate.

In this respect, although AlphaFold2 achieved a score of ~90% with regards to folding accuracy⁸¹, it has shown mixed results (often comparable to traditional homology models) when predicted protein structures are used for virtual screening of compounds^{82–84} – the next logical step in the drug discovery process. Subsequent methods, such as AlphaFold3, have improved the situation as they were explicitly designed to model molecular interactions. However, some current co-folding methods still fail to account for physical binding processes⁸⁵, making it difficult to rely on these models beyond

memorization of known structures. Open-source models in the area of co-folding have largely caught up with closed-source developments, in particular in the context of the OpenFold consortium⁸⁰. In-house developments such as a successor to AlphaFold3, IsoDDE from Iso-morphic Labs, claim superior performance in the areas of binding pocket, folding and binding energy prediction, but offer few technical details and hence do not allow independent verification of the results obtained. Diffusion models, such as those used by AlphaFold3 as well RFDiffusion (Table 3), have also attracted considerable funding, such as for the formation of Xaira, which raised US\$1 billion in its first funding round⁸⁶.

There are other areas where promising results have been reported for applications of AI, albeit the effective AI contribution to popular breakthroughs is often clinically limited. In particular, generative models that automate small-molecule design are now established approaches to improve hit finding, with published examples of experimentally validated workflows leading to higher hit rates and novel chemotype discovery^{87–89}. However, the contribution of AI technology here is predominantly in enhanced search of chemical spaces, and success is still mostly limited to ligand design rather than drug design (Box 1), due to the persistent limitations of AI models to accurately predict in vivo properties such as human bioavailability needed to guide the search with clinical relevance. Success has been claimed for some ‘AI-designed’ drugs up to phase IIa trials⁹⁰; however, attributing these successes to AI alone remains difficult as the lead-optimization steps linking the AI-generated hits to the final drug candidates are often insufficiently described or undisclosed⁹¹.

Table 3 | Selected recent developments in applications of artificial intelligence in drug discovery

Application	Key recent achievements	Challenges to achieving (clinical) impact
Small molecules		
Synthesis prediction	Successes include use of Monte Carlo Tree Search and symbolic AI ¹⁵⁷ , and publicly available software such as AlZynthFinder ¹⁵⁸ .	Limited by the low availability of negative data (reactions that are not successful under given conditions) ¹⁵⁹ , incomplete annotations (such as solvents and reaction conditions) and high redundancy of some frequent reactions, at the expense of rarer ones.
ADME prediction	'In silico-in vivo' PK models, predicting in vivo PK solely based on chemical structure as input ^{97,68,150–153} .	Chemical space coverage, biological variability, and data quality and availability pose problems ^{23,160} .
Generative models	Generative models of small molecules started with 2D methods ^{161,162} and progressed to 3D ¹⁶³ . Recent approaches can generate synthetically accessible molecules ¹⁶⁴ .	Methods have struggled to generate molecules that follow basic chemical principles such as no clashing atoms ¹⁶⁵ . Benchmarking with relevance to real-world project situations ¹⁶⁶ , and more practical benchmarking criteria have been discussed ^{167–169} . 'Failure modes' ^{170,171} (models 'cheating the scoring function') can be problematic.
Proteins, biologics and advanced therapeutic medicinal products		
Protein structure prediction	AlphaFold ⁵⁹ , AlphaFold2 ¹⁷² and AlphaFold3 ¹⁷³ have made major contributions to protein structure prediction, structure elucidation, and protein and antibody design. AlphaFold2 set the benchmark for protein structure prediction at CASP14 by a wide margin ¹⁷⁴ , whereas AlphaFold3 was able to also incorporate ligand interactions in folding.	Docking to AlphaFold models has been found to not be better than homology models in some studies ^{82–85} . AlphaFold/ML models often converge to 'what they know' (the training sets provided), resulting, for example, in incorrectly predicted disordered regions ¹⁷⁵ . For detailed discussion, see ref. 176.
Protein, peptide and antibody design	Major progress has been made with methods such as RFDiffusion ¹⁷⁷ and ProteinMPNN ¹⁷⁸ , leading to experimentally validated aims such as designing functional membrane protein analogues ¹⁷⁹ , peptide binder design from sequence ¹⁸⁰ , enzyme design ¹⁸¹ and antibody design ¹⁸² . More recent approaches include BoltzGen ¹⁸³ , Chai-2 ¹⁸⁴ and others.	Challenges include limited data and flexible binding regions. Although binding is relatively easier to predict, functional effects and developability also remain challenges ²¹ . Attempts are being made to extend beyond designing structure towards designing and tuning function and interactions.
Cell and gene therapy	Current approaches include the use of models for efficacy and safety ^{185,186} , and especially in manufacturing ^{187,188} .	The relative immaturity of the field means there is often little data available, and standard processes for experimental work and data generation have not been defined.
High-dimensional functional assay development		
High-content imaging	ML methods applied to Cell Painting or microscopy images have been used to train functional models used for phenotypic drug and target discovery, disease modelling and target or drug validation efforts ^{105,110,124,185,186,189,190} .	Recent advances have not yet reached clinical validation stages. Strong data generation and quality controls are needed to ensure that models are trained on biologically relevant data. Ideally, models will be trained across samples with sufficient biological diversity to identify generalizable phenotypic differences associated with drug responses or diseases.
Temporal, single-cell and spatial transcriptomics	These approaches have enabled characterization of biological systems (including disease models and the effects of compounds on them) at a finer resolution than before ^{102,122} .	A transition from generating high-dimensional data to identifying variables for clinically relevant decision making is key for the impact of such data.
Technology		
Deep learning	Deep learning approaches, including CNNs ¹⁹¹ , RNNs ¹⁹² and transformers have progressed tremendously over the last decade. Vision transformers have been used to navigate complex imaging data, with advances in image segmentation ¹⁹³ and phenotypic modelling ¹⁹⁴ .	Clinical impact has been relatively modest so far owing to the limited availability of data types that enable clinical translation ²⁹ . Nevertheless, deep learning has improved the predictivity of some (in particular physicochemical) end points gradually (and sometimes considerably ¹⁹), especially where a large amount of data is available.
LLMs	LLMs have attracted much attention following the release of GPT-3 in 2020 ¹⁹⁵ , but it should be emphasized that they solely process patterns in language on a statistical basis, rather than 'reason' ¹⁹⁶ . LLMs have been used for de novo design ¹⁹⁷ and to interface with dedicated domain-specific tools, such as for chemistry in ChemCrow ¹⁹⁸ .	Although LLMs can help automate processes and summarize information, the crucial factor in drug discovery is often making better decisions in an information-sparse environment, where the added value of LLMs remains to be seen. Agentic LLMs for different 'tasks' in a drug discovery process have been described ^{199,200} , but solely achieving high scores on such tasks may not be sufficient to improve drug discovery processes, given that the tasks are underspecified and do not represent the real-world situation ⁴⁷ .
Translation to the clinic		
RWE and using LLMs for diagnosis	EHRs can be analysed for purposes such as adverse event detection ²⁰¹ , biomarker discovery ²⁰² and improved detection of underdiagnosed diseases, particularly rare diseases ²⁰³ .	EHR data pose issues with insufficient completeness, the use of ontologies and vocabularies that do not capture all necessary information, and biases ^{204,205} . Although LLMs have been used to analyse patient information (for example, for diagnosis), such applications require a fit-for-purpose assessment, and positive ²⁰⁶ , unsatisfactory ²⁰⁷ and mixed ²⁰⁸ results have been reported. In particular, the data available in a real-world situation often does not resemble the more complete data that has been used in some LLM studies.

Table 3 (continued) | Selected recent developments in applications of artificial intelligence in drug discovery

Application	Key recent achievements	Challenges to achieving (clinical) impact
Translation to the clinic (continued)		
Clinical trial design and recruitment	Algorithms for clinical trial design can assist with cohort composition (and trial design), patient recruitment, and retention and monitoring ²⁰⁹ . Even minor improvements related to clinical trial conduct can be highly impactful, given the high cost of recruitment ²¹⁰ and how frequently failure to recruit is cited as the reason for clinical trial termination ²¹¹ . Virtual control groups for clinical trials have also attracted interest ²¹² .	Clinical trial design has benefited from the application of computational algorithms ²⁰⁹ . This is enabled by the provision of suitably structured patient data.
Clinical outcome prediction	Various implementations of clinical outcome prediction approaches have been reported ^{213–215} , with claims of accuracy of around 80%.	Clinical outcome prediction has direct clinical relevance and can be useful. Generally, predictions will be better in areas where previous data are available. Also, one needs to be aware of ‘survivor biases’ and reinforcement loops in previously successful areas, which could lead to decreased biological novelty of future studies.

ADME, absorption, distribution, metabolism and excretion; CNN, convolutional neural network; EHR, electronic healthcare record; LLM, large language model; ML, machine learning; PK, pharmacokinetics; RNN, recurrent neural networks; RWE, real-world evidence.

In some cases where sufficient data are available and where descriptors can capture underlying trends (such as in case of logD and other physicochemical properties¹⁹), the application of AI approaches such as deep learning in drug discovery seems to have contributed to improved predictions. The picture is mixed, with some successes reported (see, for instance, refs. 92,93), but also counterexamples (see, for instance, refs. 79,94–96).

Overall, approaches and use cases are very dependent on the modality of interest (Box 3), with small-molecule and sequence-based AI models often being focused on target binding and affinity aspects of drug discovery, whereas in areas such as cell therapy, current AI applications are more often concentrated on manufacturing and production. What is shared across different applications is that the integration of data, models and use cases deserves more attention for real-world impact in the future. This consideration also motivates the inclusion of high-dimensional biological assays as a requirement to make an impact. For in vivo-relevant decision making, it is necessary to understand where biologically meaningful signals reside, generate data in systems that reflect in vivo biology, build appropriate computational models and validate them in a use-case-relevant manner. This is in contrast to situations where data are generated primarily through ‘tech push’ rather than ‘science pull’, and then later repurposed for new applications.

Where do we go from here?

First, we need to identify where the greatest gaps exist between current data used in drug discovery and their predictivity for outcomes in clinical development. In other words, we have to make decisions about what needs to be measured to bridge the translational gap and to generate and categorize data accordingly, which we discuss further below.

Second, we need focused efforts to iterate on AI model design and validation in a use-case-relevant manner, including technical aspects such as reproducibility and computation modelling approach. In this second phase, the process can be productionized and scaled, increasing speed and reducing costs, while also being refined to continually generate data optimized for AI analysis and subsequent decision making. In short, it is more important to do the right thing in the first place than to do things, technically, right (if these are the wrong things to do). These two steps are visualized in Fig. 6a in the form of iterative as well as subsequent circles.

In our view, there is currently less of an observable focus on the first phase, which in many cases may be due to shortage of time and resources or protection of industrial knowledge, but sometimes also due to an over-reliance on technology as a driver. We should generate the right data for a GIVEN purpose, especially when consuming considerable resources. However, the second phase also lacks attention, and some algorithms that have been around for a long time have only been scaled, productionized and rolled out to project teams in pharmaceutical companies relatively recently. This is frequently because of integration barriers: siloed data, complex legacy systems, coordination across multiple stakeholders and the need to train end users^{97,98}. As a result, ‘AI in drug discovery’ can mean two very different things: advanced, rapidly developed methods that lack thorough validation, as well as mature, established methods that have only just become practical to use at scale in production environments. What is common to both situations though is that the translation from ‘AI’ to ‘drug discovery’ has not yet been observed in an efficient and sufficiently durable manner to lead to clinical impact.

When it is clear that efforts should prioritize clinical translation (for example, the efficacy of targeting an untested pathway or safety of a new drug modality), then the predictivity of any model for outcomes in clinical development must come first^{20,29,57}. Although the specifics of what is ‘right to do’ will depend on the context, the value of any computational model will ultimately be determined by two characteristics: its broad applicability across project decision points and its ability to improve clinical translation. This is visualized in equation 1 in a simplified form (aspects such as project value are left out, as they are often unknown).

$$\text{Model value} = \text{Improvement in clinical success rate} \times \text{Project applicability domain.} \quad (1)$$

The value of a predictive model increases with the impact on the clinical success of projects where the model is applied (as opposed to only earlier proxy measures, such as hit rates). The value of the model also increases with the breadth of the project applicability domain; for example, models for safety assessment or pharmacokinetics are more generally applicable across projects, whereas target-specific or disease-specific models will be limited to their particular use case. Another aspect of the applicability domain is defined by the data on which a given model is based, such as the chemical space or modalities it covers.

Identifying what to do

Here, we share our perspective on the areas where we believe AI developers and practitioners, as well as organizations and companies, should focus more strongly to increase the likelihood of successfully applying AI in drug discovery. The generation and selection of AI-optimized data that are predictive of human efficacy and safety is critical for successful drug development, and ideally this can be achieved via increasingly cheaper and more informative proxy systems. Although this strategy has been proposed before²³, the landscape has since evolved: the availability of both more generic and more tailored complex biological systems, together with advances in automation and data analysis, now enables approaches that were not feasible at the time of that earlier work.

As a generic approach to characterizing compound action and disease perturbation in biological systems, we can develop *in vitro* or *ex vivo* assays that utilize high-dimensional functional readouts such as Cell Painting^{99–101} or single-cell RNA sequencing, and then use ML to discriminate between physiological cellular states¹⁰². Also ‘biological perturbation maps’ have been constructed based on a combination of transcriptomics and cell morphology data, linking gene perturbations to biological effects on a large scale (for example, on the protein complex or pathway level¹⁰³), thereby showing some success in deriving the biological (here pathway-level) impact from cost-effective assay systems.

ML models built on top of high-dimensional biological assays could be more representative of human biology, enabling preclinical development that translates better to clinical outcomes. Examples include phenotypic screening using ML to classify inflammasome-perturbed cellular states (a higher-dimensional readout as opposed to the typical single-readout of IL-1 β ¹⁰⁴), ML classification to predict mitochondrial toxicity

phenotypes of proteolysis-targeting chimeras (PROTACS) from Cell Painting data¹⁰⁵, or assay data alone used to map gene signatures for repurposing¹⁰⁶. As an extension, when such biological assays for phenotypic screening are performed in human primary cells¹⁰⁷ (ideally from many donors) rather than immortalized cell lines or animal models that lack genetic diversity, we can introduce more human variability into preclinical testing and identify drugs that are more likely to be generally effective across diverse human populations¹⁰⁸. For such readouts to be useful, experimental parameters and their predictive value for specific efficacy or safety end points must be established largely empirically. In the long term, approaches such as ‘virtual cells’ may be able to connect perturbation, whether from disease or drug perturbations, to phenotype-level outcomes¹⁰⁹.

Another extension for disease-relevant biological assays is to reflect that cell-type-specific vulnerability is often central to disease aetiology. In neurodegeneration, this is exemplified by the selective loss of motor neurons in amyotrophic lateral sclerosis (ALS) and dopaminergic neurons in Parkinson disease¹¹⁰. In Alzheimer disease, by contrast, non-cell autonomous effects from dysregulation in microglia, astrocytes and other cell types are likely causal drivers¹¹¹. Primary samples, however, are inaccessible and simplistic model systems that do not capture relevant physiology or pathophysiology. Here, human induced pluripotent stem (hiPS) cell-based biological models capture uniquely human neuronal features absent in animals and have emerged as the most promising systems to study disorders such as ALS, Parkinson disease and Alzheimer disease, with early evidence of impact in both disease understanding and drug discovery¹¹², representing a suitable *in vitro* model and data-generation system. For example, hiPS cell-derived motor neurons from patients with ALS have been used to identify

Box 3 | Differences between data types

We have emphasized the differences between life sciences data and other data types such as photographic images and voice audio. It is also important to acknowledge the differences between subtypes of life science data, here somewhat simplified to data in the biological, chemical and physical domains. It is critical to recognize that approaches that work in one domain only rarely transfer to another, given that the data are intrinsically very different.

Biological entities can often be represented as a one-dimensional sequence of a small and well-defined alphabet of possible building blocks, where there is an underlying structure between elements on the one hand, and an overarching selection principle on the other hand. These aspects make both the representation of the entity easier and limit the size of the model domain (for a discussion of underlying principles, see ref. 221). For example, in the case of protein sequences, the alphabet building blocks are the 20 naturally occurring amino acids; the underlying relationship between amino acids is captured in substitution matrices (such as the BLOSUM matrix and successors), and this is linked to evolution as a functional selection criterion of the overall protein. This underlying, low-dimensional structure of biological sequence space, and its mapping to 3D structural space, is a fundamental reason for the success of AlphaFold in this problem domain.

Chemical data are completely different because there is no underlying one-dimensional representation of chemical structures

that is equivalent to that for proteins. Although one-dimensional notations of small molecules (such as SMILES notations) exist, they are solely a notation of structure and do not correspond to the physical sequence of atoms in the molecule. The multidimensionality of chemical entities results in much greater complexity. In practical terms, the chemical space even for small molecules (<500 Da) is much larger (comprising $\sim 10^{60}$ ‘small’ molecules³⁹) than that of one-dimensional biological sequences, and chemical datasets are also biased — from synthesis bias, to analogue/project bias and publication bias. In small-molecule chemical space, we are only ever able to shed light on small areas in a biased way, and in most cases we are unable to extrapolate between different ‘galaxies’ (Fig. 3). This means that in this context we are not able to calculate aspects such as significance or absolute likelihoods in resulting models in any meaningful (unbiased) way.

By contrast, physical data can be understood based on underlying principles relating structure to measurable end points, which are formalized into theories and equations. For example, we are able to calculate band gaps for absorption of light, or the interaction of orbitals between a catalyst and a substrate in a chemical reaction. This is the advantage of materials or catalyst design — we understand the system from first principles, leading to the ability to fill gaps in data with calculations in many cases.

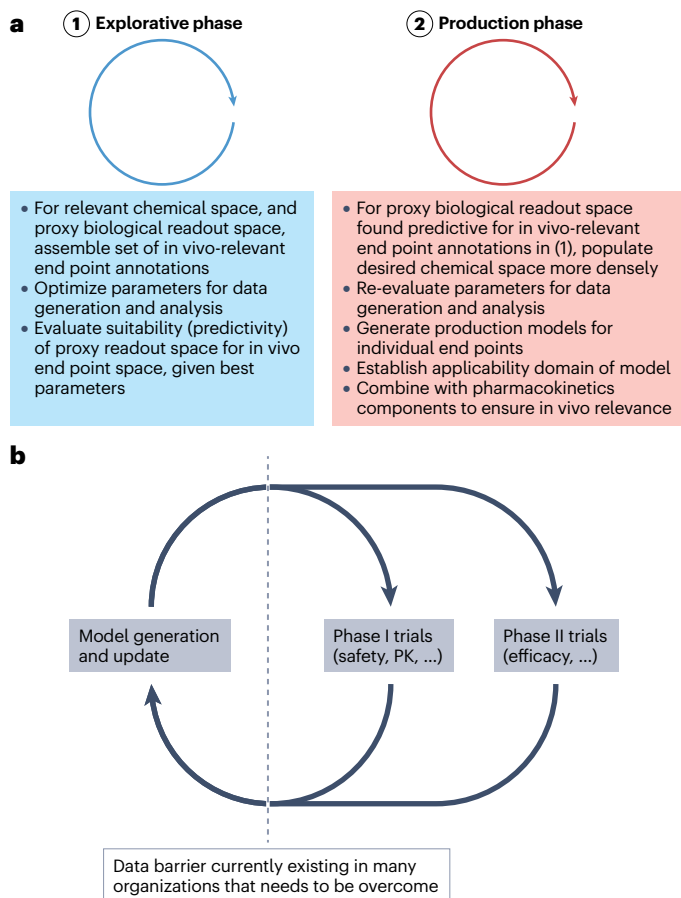


Fig. 6 | The generation of suitable data for predictive artificial intelligence models in drug discovery. **a**, Data generation requires two phases to ensure suitability for a given use case. First, it requires understanding which proxy data need to be generated to ensure predictivity of a proxy assay for a clinically relevant end point. This is a more explorative phase, which includes the generation of smaller-scale datasets using different set ups of a proxy screening system. Second, in the production phase, data generation needs to be scaled up. With suitable experimental design, the model needs to be tuned for and validated on this larger dataset, its applicability domain needs to be established and finally the model needs to be integrated with pharmacokinetics models to ensure translation to the clinical situation, which takes exposure at the site of action into account. **b**, To improve the predictivity of models with clinical relevance, artificial intelligence models predictive for preclinical end points need to learn from clinical observations. For example, a model for microsomal stability of compounds has a certain predictivity for the clinical situation of estimating absorption, distribution, metabolism and excretion parameters, and the strength of this information then determines the suitability of a predictive model for microsomal stability in the preclinical context. This type of learning across the preclinical and clinical parts of an organization is often challenging in practice owing to organizational and technical reasons (clinical data have different data access requirements, and it is stored in different databases in different parts of an organization). PK, pharmacokinetics.

disease-relevant phenotypes and to test candidate compounds such as ropinirole and bosutinib in vitro, which has already led to clinical candidates¹¹³. iPSC-derived systems of motor neurons have also been used in combination with high-content imaging and deep learning to

capture morphological signatures of eight familial ALS mutations, and hence to incorporate clinically relevant biology in practically manageable test (and screening) systems¹¹⁴.

These applications, although still limited and largely focused on existing hypotheses, highlight the principle that predictive validity in hiPS cell models matters. With regulatory momentum from initiatives such as the FDA Modernization Act 2.0, the field is increasingly incentivized to adopt this type of more human-relevant, mechanism-rich platform earlier in discovery¹¹⁵. Cell Painting assays, which as noted above are in principle of more generic nature, can also be tailored to more specific use cases, as shown by a recent study deriving a fibroblast-based model for intestinal fibrosis¹¹⁶, identifying both hits in a relevant model system and modes of action simultaneously. Other predictive biological model systems, such as organ-on-a-chip¹¹⁷ or organoid models^{118,119}, may also offer complementary or alternative avenues for future development.

Beyond human disease models, pathogens such as bacteria can be used as screening systems for generating data to train ML models. This is illustrated by a recent study using *Escherichia coli* screening data to train a deep learning model that was able to identify a diverse group of antibacterial hits with high dissimilarity to known antibiotics¹²⁰.

For some modalities, parameters relevant to clinical translation can increasingly be modelled effectively. For example, although high affinity for antibodies is not sufficient for efficacy per se, it is a requirement for it, and antibody design has made great strides in recent years (see Table 3 for examples and references). Likewise, developability of antibodies (such as thermostability, expression and immunogenicity) is an area where ML methods have progressed recently. Although no model is able to predict all such end points yet, some developability end points can predicted better than random, and fine tuning models seems to greatly improve model performance, and hence targeted data generation can also aid translation here¹²¹.

To get the full value out of AI and ML for analysing clinically relevant biological data, we need models of human biology – or of other measurable assays related to clinical success, such as heterogeneous co-culture or microfluidic systems – that are complex enough, large scale enough and detailed enough to accurately capture how biological systems work¹²². This means that systems biology and functional genomics need to evolve: it is no longer enough to simply generate huge datasets, we need data that actually reveal mechanisms and matter clinically. Current developments in single-cell and spatial omics are progressing in the right direction¹²² – if we operate in biologically relevant systems and consider which data to generate for which use case.

If we can perform genome engineering at scale in the right kinds of human cells, or in sophisticated laboratory models that mimic real tissue interactions, we can build experimental systems where the genetic change (perturbation), the biological context and the resulting phenotype all line up. This creates a ‘causal genetics’ framework¹²³, where we can directly test disease-related hypotheses and link mechanistic causes to phenotypic effects¹²⁴. These tests could involve knocking out genes, precisely editing protein-coding regions, engineering non-coding variants implicated by genome-wide association studies, and even exploring therapeutic approaches such as antisense oligonucleotides, small molecules or biologics.

However, these perturbations only become meaningful if they are made in relevant biological systems and measured with the right multi-omic tools¹²². ‘Relevant biological systems’ here also need to evolve beyond cell monocultures – and, where relevant for the disease context, towards co-culture systems to capture cellular crosstalk, both

between native human cells, as well as between diseased cells and their environment, such as in the case of understanding interactions between cancer cells and the tumour microenvironment. Building all these capabilities together would let us create large, context-rich datasets that accurately depict states of human health and disease. Those datasets are exactly what AI and ML models need in order to learn accurate, clinically relevant predictive relationships. Ultimately, this shifts human functional genomics from focusing on modelling individual patients' diseases to becoming an integrated engine for causal discovery and translational prediction, aimed at improving the success rate of drug discovery. Moreover, such *in silico* systems-biology assays can then be used to evaluate therapeutic candidates by modelling how compounds with known or predicted polypharmacological activity influence interconnected pathways, enabling screening based on predicted network-level responses rather than single-target effects.

Integrating preclinical and clinical data for discovery

Pharmaceutical companies often have barriers between their pre-clinical and clinical functions, owing to rigid data protection policies established in the clinical domain. Learning from clinical outcomes, however, is crucial to determine which data are predictive for clinical end points (Fig. 6b). In an indication-specific manner, iterative data generation and feedback between the early stages of drug discovery and phase I trials can be used to train models on end points that are associated with clinical tolerability, toxicity and pharmacokinetics. Similarly, outcome data from phase IIb proof-of-concept trials can be used to iterate on target validation and compound efficacy predictability models. By including negative or suboptimal clinical outcomes data too, such models may not only improve preclinical translation into clinical successes, but could also help mitigate the confirmation biases that drive clinical pipelines.

Model validation in a project-centric and use-case-centric manner

Current model validation is, in some cases, model centric, and performance metrics are not always tailored to the eventual model use case. To make AI useful in drug discovery, model deployment and validation need to be tightly coupled to specific project contexts and decision points. In practice, this means evaluating models within the workflows they are intended to support, accounting for realistic data availability and update cycles, and selecting performance metrics that reflect the actual decisions being made, along with their quantitative trade-offs, which can differ substantially between early discovery and later development stages. It also requires considering which experimental follow-ups are feasible and informative in a given setting, rather than assuming 'idealized validation scenarios' (see above). Approaches such as embedding AI models into design–make–test–analyse (DMTA) cycles are progressing in this direction, and necessarily need to consider such aspects in order to be successful¹²⁵. In addition, companies such as Ikto¹²⁶, XtalPi¹²⁷, Insilico Medicine¹²⁸, Genentech, Tempus and many others are implementing AI-integrated DMTA cycles, including those that explicitly incorporate data from patients and complex biology models such as organoids. What remains key is that the selection criterion used is predictive for the later human situation, such as evaluating compound activity using human-relevant assays, so that the advantage of automation leads to improvements in quality, and not just quantity. With this Perspective, we are aiming to assist a change in thinking about what a 'validated model' is and how to 'perform validation', in areas from medicine⁷² to the material sciences⁷³, and with explicit consideration of

data quality and its impact on model performance^{77,78,129}. Drug discovery has been lagging but is slowly also adopting such methods of more relevant performance assessment^{42,48,65,130}.

Possible paths in practice

One practical route to enabling AI-based drug discovery is the formation of large, industry-spanning consortia that include the pharmaceutical sector and adjacent industries such as agrochemistry and consumer goods. Such consortia are motivated by scale: the size of chemical space to be covered, the number of proxy end points and assays that can be performed, and the number of *in vivo*-relevant end points those assays can be linked to, cannot be covered by one company or entity alone.

Although some consortia, such as the IMI consortia¹³¹, have been formed to this end before, these sometimes fell short because experimental design and context-dependent data generation were not treated as a first-class concern. Data were frequently pooled retrospectively, having been generated for different purposes and under heterogeneous conditions. For example, clearance data might have been pooled across companies despite differences in assay format, species or experimental thresholds, implicitly assuming equivalence where none did exist. This affected the outcomes in two ways: first, chemical space coverage was largely taken as a given, rather than being deliberately designed to support fit-for-purpose predictive modelling, exacerbating bias and limiting generalizability. Second, assay conditions were not harmonized across companies, leading to data that have been to some (often unknown) degree inconsistently measured. Therefore, to improve the predictivity of AI models in the future, these two points need addressing by companies (which can define needs and consider experimental design to generate suitable data for harmonized ML model training), supported by cross-industry and government funding (which could be organized similarly to the IMI consortia).

An alternative path is the emergence of single organizations, whether industrial, venture backed or publicly funded, that generate large, purpose-built datasets once an exploratory phase has clarified the relevant questions. The advantage of this approach is that the organizational set up is probably easier. The primary challenge is achieving the initial momentum and funding required to operate at sufficient scale. For example, in the area of ADMET, the OpenADMET consortium (funded via ARPA-H with US\$30 million), is generating data for off targets that have been linked to clinical liabilities (as well as other data). This type of data, generated with ML models in mind, and for relevant end points, will be of great value for its end users. Efforts such as LIGAND-AI and the Virtual Cell Pharmacology Initiative cover different types of data, namely ligand–target interaction data and compound DRUG-seq readouts, but both efforts also aim at large-scale and consistent datasets that will be made publicly available. The first two examples listed here are publicly funded, whereas the third is from a biotech company (Ginkgo Bioworks), showing that both avenues are possible, for different reasons and depending on the particular situation.

To summarize, we, as a field, need to continue moving from a technology-focused view to a science-focused view on what we should, and need, to do, to really take advantage of the experimental and computational capabilities we have. Second, we need more focus on approaches that are process validated, that scale and that consistently get used by experimental scientists, which requires care when developing and implementing models, as well as aspects such as interface design, education and cultural change in an organization.

Human psychology and extrinsic motivators

‘AI in drug discovery’ operates in a broader societal context. Consequently, extrinsic motivators have a profound impact on how the technology is deployed and how the field advances. In biopharma R&D, a critical contextual factor is the increasing cost required to achieve an approved drug (‘Eroom’s Law’¹³²), exerting a pressure on pharmaceutical companies to change processes. At the same time, ‘AI first’ companies are often launched with extravagant promises, escalating the pressure faced by both the company and its investors to go all in on those promises, which can adversely impact decisions¹³³. For some decision makers in pharma companies, the impact of emotions and biases in decision making might be underappreciated^{134–136}. Scientific publishing (and academic research), in its pursuit of ‘high impact factors’ (and ‘impact’ and obtaining grants and promotions), also contributes to a distortion of scientific progress. This can result in the publication of substandard work (Peplow discusses one such example from materials discovery¹³⁷) and may help explain why ‘high-impact’ journals are seeing an uptick in retractions¹³⁸. In the context of the topic discussed here, models that ‘numerically push the state-of-the-art’ on toy datasets are published and receive a lot of attention, whereas their concrete applicability to, and impact on, real-world processes may be close to non-existent or be overlooked. In combination with the insufficient attempts to truly validate models (such as through prospective validation and translation to a decision-making process), an illusion of progress is created; however, impact ‘at the coalface’ is in many cases overstated.

In the bigger context, and partly out of necessity, much of biological research is based on the reductionist idea of disassembling a system into components (such as genes and proteins) to understand the impact of these and identify intervention points¹³⁹. The implicit assumption is that modulating a single component will produce the desired effect once the system is re-embedded in its full biological context. However, this assumption often fails owing to emergent effects, such as pathway redundancy, feedback loops and cell–cell interactions¹⁴⁰, contributing to high clinical failure rates driven by lack of efficacy. A common example is target inhibition that shows clear pathway modulation in vitro or in animal models but fails to translate into meaningful clinical benefit once compensatory mechanisms are engaged in humans¹⁴¹.

In response, multiomics approaches have emerged to characterize biological systems across multiple organizational levels, giving rise to systems biology¹⁴². The central challenge then shifts to selecting the ‘right variables’, measured in the ‘right biological system’ and at the ‘right time’ with respect to both disease progression and intervention. Even when such data are available, extracting actionable conclusions from these very high-dimensional datasets¹⁴³ remains difficult. In most practical settings, the number of measured variables far exceeds the number of informative samples, placing strong limits on purely data-driven inference. Integrating data with prior biological knowledge – for example, with Bayesian or causal modelling approaches¹²³, or approaches such as Mendelian randomization¹⁴⁴ – is therefore likely to be a necessary component of any viable solution.

Conclusion

The relevant success metrics for AI in drug discovery need to be numbers that matter, rather than contrived proxy measures that primarily optimize for scientific publications rather than actual R&D progress. Ultimately, what truly matters is increasing clinical approval rates, as well as novelty in both modality and mode of action space. We hope that

this Perspective contributes to a realistic view on current approaches, and helps clarify the conditions under which AI can deliver tangible impact in the future.

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Competing interests

A.B. is a shareholder or options holder of or consultant with Healx, Pharmedica Therapeutics, Pangea Bio, Cresset Group, Evolvus, Harmonic Discovery, Ignota Labs, Turbine, VALID, Daimeidex, EtioMap, Apheris, AstraZeneca and Novartis. M.C.T. has no competing interests to declare. J.W.S. is director and shareholder of JW Scannell Analytics, director and shareholder of Ethers Pharmaceuticals, holds equity options in Ochro Bio and has an advisory role with Hiro Capital. D.A.S. has no competing interests to declare. G.M.G. is a shareholder in AstraZeneca, Vertex and Nvidia. J.G.G. has no competing interests to declare. L.-L.P. has no competing interests to declare. R.D.J. is CEO and cofounder of Powerhouse Biology and owns shares in the company. K.H. is an employee of Meiji Seika Pharma Co. and is a shareholder of Teijin Pharma. M.H. is an employee of Tanabe Pharma Corporation. S.S. is an employee of Human Chemical Company. M.M. has no competing interests to declare. M.F.S. is an employee and shareholder of biot.ai. T.A. is CEO and cofounder of VALID and owns shares in the company. F.G. has no competing interests to declare. I.C.-C. has no competing interests to declare.

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